STPM Physics · Semester 2

FORMULA LIST

Electricity & Magnetism · Chapters 13–19. 181 formulas with symbols, units and conditions. Numbered chapter by chapter, taken from the PhysiQ notes.

CHAPTER 13 — ELECTROSTATICS

13.1 Coulomb's Law

1	**Coulomb's law (vacuum)**	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2} = \frac{kQ_1Q_2}{r^2}$	F = electrostatic force (N); Q_1, Q_2 = point charges (C); r = separation (m); $k = \frac{1}{4\pi\varepsilon_0} = 8.99\times10^9$ N m^2 C^{-2}; $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
2	**Coulomb's law in a medium**	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0\varepsilon_r r^2}$	ε_r = relative permittivity of the medium (no unit); other symbols as above
3	**Force in an electric field**	$F = qE$	F = force (N); q = charge (C); E = electric field strength (N C^{-1} or V m^{-1})
4	**Acceleration of a charge**	$a = F/m$	a = acceleration (m s^{-2}); F = force (N); m = mass (kg)
5	**Charged spheres on threads**	$\tan\theta = \frac{F}{mg}$	Two small identical charged spheres on threads; θ = angle of thread to the vertical; F = horizontal electric force (N), balanced by the tension and the weight mg; m = mass (kg); g = acceleration of free fall (m s^{-2})

13.2 Electric Field

6	**Electric field strength**	$E = \frac{F}{q}$	E = electric field strength (N C^{-1} = V m^{-1}); F = force on the test charge (N); q = test charge (C)
7	**Field of a point charge**	$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{kQ}{r^2}$	Point charge, or outside a charged sphere; Q = charge (C); r = distance from the centre (m)
8	**Field between parallel plates**	$E = \frac{V}{d}$	V = potential difference (V); d = plate separation (m)

9	**Field and potential gradient**	$E = -\frac{dV}{dr}$	$\frac{dV}{dr}$ = potential gradient (V m^{-1}); the field points in the direction of decreasing potential
10	**Force and acceleration of a charge**	$F = qE$; $a = \frac{qE}{m}$	F = force (N); a = acceleration (m s^{-2}); q = charge (C); m = mass (kg)
11	**Charge starting from rest**	$v^2 = 2as$	Charge starting from rest with uniform acceleration; v = final speed (m s^{-1}); s = distance (m)
12	**Charge entering perpendicular to field**	Parabolic path	The path is a parabola, with constant velocity component along the plates
13	**Field near a charged conducting plate**	$E = \frac{\sigma}{\varepsilon_0}$	Near a large charged conducting plate; σ = surface charge density (C m^{-2}); ε_0 = permittivity of free space (F m^{-1})

13.3 Gauss's Law

14	**Gauss's law**	$\Phi = \oint E \cdot dA = \frac{\Sigma Q}{\varepsilon_0}$	Φ = electric flux (N m^2 C^{-1} = V m); E = electric field strength (N C^{-1}); A = area (m^2); ΣQ = algebraic sum of the enclosed charges (C); ε_0 = permittivity of free space (F m^{-1})
15	**Field of a point charge or sphere**	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$	Valid for r ≥ radius R; Q = charge (C); r = distance from the centre (m)
16	**Field of an infinite line charge**	$E = \frac{\lambda}{2\pi\varepsilon_0 r}$	λ = charge per unit length (C m^{-1}); r = perpendicular distance from the line (m)
17	**Field of an infinite plane sheet**	$E = \frac{\sigma}{2\varepsilon_0}$	σ = charge per unit area (C m^{-2})
18	**Field at a conductor surface**	$E = \frac{\sigma}{\varepsilon_0}$	Just outside the surface of a conductor; σ = charge per unit area (C m^{-2})

13.4 Electric Potential

19	**Potential of a point charge**	$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{kQ}{r}$	V = electric potential (V); Q = charge (C); r = distance from a point charge, or from the centre of a charged sphere for r ≥ R (m)
20	**Work done by an external force**	$W = q(V_B - V_A) = q\Delta V$	Work done by an external force in moving charge q from A to B (J); q = charge (C); V_A, V_B = potentials at A and B (V)
21	**Work done by the field**	$W = q(V_A - V_B)$	Work done by the field in moving charge q from A to B (J)

22	**Electric potential energy**	$U = qV$	U = potential energy (J); q = charge (C); V = potential (V)
23	**Potential energy of two point charges**	$U = \frac{kQ_1Q_2}{r}$	Q_1, Q_2 = point charges (C); r = separation (m); $k = \frac{1}{4\pi\varepsilon_0}$
24	**Field and potential**	$E = \frac{V}{d}$; $E = -\frac{dV}{dr}$	$E = \frac{V}{d}$ for a uniform field; $E = -\frac{dV}{dr}$ in general; E = field strength (V m^{-1}); d = separation (m)
25	**Charge accelerated through p.d.**	$\frac{1}{2}mv^2 = qV$	Charge accelerated from rest through potential difference V; m = mass (kg); v = speed (m s^{-1}); q = charge (C); V = potential difference (V)
26	**Charged oil drop at rest**	$\frac{qV}{d} = mg$	Charged oil drop held at rest between plates; q = charge (C); V = p.d. (V); d = plate separation (m); m = mass (kg); g = acceleration of free fall (m s^{-2})

CHAPTER 14 — CAPACITORS

14.1 Capacitance

1	**Capacitance**	$C = \frac{Q}{V}$	C = capacitance (F); Q = charge on one plate (C); V = potential difference across the plates (V)
2	**Capacitance of an isolated sphere**	$C = 4\pi\varepsilon_0 R$	Isolated conducting sphere; R = radius (m); ε_0 = permittivity of free space (F m^{-1})
3	**Parallel plate capacitor (air)**	$C = \frac{\varepsilon_0 A}{d}$	A = area of one plate (m^2); d = plate separation (m)
4	**Parallel plate capacitor (dielectric)**	$C = \varepsilon_0\varepsilon_r\frac{A}{d}$	ε_r = relative permittivity of the dielectric (no unit)
5	**Unit of capacitance**	$F = C\,V^{-1} = A^2\,s^4\,kg^{-1}\,m^{-2}$	Typical values are μF (10^{-6} F), nF (10^{-9} F) and pF (10^{-12} F)

14.2 Parallel Plate Capacitors

6	**Parallel plate capacitance (air)**	$C = \frac{\varepsilon_0 A}{d}$	C = capacitance (F); A = area of one plate (m^2); d = plate separation (m); $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
7	**Capacitance with dielectric**	$C = \varepsilon_0\varepsilon_r\frac{A}{d}$	Dielectric completely filling the gap; ε_r = relative permittivity (no unit)

8	**Field between the plates**	$E = \frac{V}{d} = \frac{Q}{\varepsilon_0 A}$	E = electric field strength (V m^{-1}); V = potential difference (V); Q = charge (C)
9	**Layers in series**	$\frac{1}{C} = \frac{d_1}{\varepsilon_0 A} + \frac{d_2}{\varepsilon_0 \varepsilon_r A}$	d_1, d_2 = layer thicknesses (m); ε_r = relative permittivity of the second layer
10	**Stored energy**	$U = \frac{1}{2}CV^2$	U = energy stored (J); C = capacitance (F); V = potential difference (V)

14.3 Dielectrics

11	**Capacitance with dielectric**	$C = \varepsilon_r C_0 = \varepsilon_0 \varepsilon_r \frac{A}{d}$	C = capacitance (F); C_0 = capacitance without dielectric (F); ε_r = relative permittivity (no unit); A = plate area (m^2); d = plate separation (m)
12	**Isolated capacitor (Q constant): field and p.d.**	$E = \frac{E_0}{\varepsilon_r}$; $V = \frac{V_0}{\varepsilon_r}$	Capacitor isolated at constant Q; E_0, V_0 = field strength (V m^{-1}) and p.d. (V) without dielectric
13	**Connected to battery (V constant)**	$Q = \varepsilon_r Q_0$; $U = \frac{1}{2}\varepsilon_r C_0 V^2 = \varepsilon_r U_0$; $E = \frac{V}{d}$ unchanged	Battery connected, V constant; Q_0 = charge without dielectric (C); U_0 = energy without dielectric (J); U = energy (J)
14	**Isolated capacitor: stored energy**	$U = \frac{Q^2}{2C} = \frac{U_0}{\varepsilon_r}$	Isolated, Q constant; U = energy (J); Q = charge (C); C = capacitance (F)
15	**Maximum voltage**	$V_{max} = E_{max}\, d$	V_{max} = maximum safe voltage (V); E_{max} = breakdown field strength (V m^{-1}); d = plate separation (m)

14.4 Capacitors in Series and in Parallel

16	**Capacitors in parallel**	$C = C_1 + C_2 + C_3 + \ldots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
17	**Capacitors in series**	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
18	**Two capacitors in series**	$C = \frac{C_1 C_2}{C_1 + C_2}$	C_1, C_2 = capacitances (F)
19	**Series charge and voltage sharing**	$Q = CV$; $V_1 = \frac{VC_2}{C_1 + C_2}$; $V_2 = \frac{VC_1}{C_1 + C_2}$	Q = charge on each capacitor (C); V = total p.d. (V); V_1, V_2 = p.d. across C_1 and C_2 (V)
20	**n identical capacitors**	Parallel: nC; series: $\frac{C}{n}$	n = number of identical capacitors, each of capacitance C (F)

21	**Total stored energy**	$U_{total} = \frac{1}{2}C_{total}V^2$	U_{total} = energy of the whole arrangement (J); C_{total} = total capacitance (F); V = p.d. across the arrangement (V)

14.5 Energy Stored in a Charged Capacitor

22	**Energy stored in a capacitor**	$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	U = energy (J); Q = charge (C); V = potential difference (V); C = capacitance (F)
23	**Energy density**	$u = U/(Ad) = \frac{1}{2}\varepsilon_0 E^2$	u = energy density (J m^{-3}); A = plate area (m^2); d = plate separation (m); E = field strength (V m^{-1}); ε_0 = permittivity of free space (F m^{-1})
24	**Energy density in a dielectric**	$u = \frac{1}{2}\varepsilon_0\varepsilon_r E^2$	ε_r = relative permittivity (no unit)
25	**Work done by the battery**	$W = QV = CV^2$	W = work done by the battery (J); Q = charge (C); V = p.d. (V); C = capacitance (F)
26	**Parallel sharing**	$V = \frac{Q_{total}}{C_1 + C_2}$	V = common p.d. after sharing (V); Q_{total} = total charge (C); C_1, C_2 = capacitances (F)

14.6 Charging and Discharging of a Capacitor

27	**Discharging a capacitor**	$Q = Q_0e^{-t/RC}$; $V = V_0e^{-t/RC}$; $I = I_0e^{-t/RC}$	Q, V, I = charge (C), p.d. (V), current (A) at time t (s); Q_0, V_0, I_0 = initial values; $I_0 = \frac{V_0}{R}$; R = resistance (Ω); C = capacitance (F)
28	**Charging a capacitor**	$Q = Q_0(1 - e^{-t/RC})$; $V = V_0(1 - e^{-t/RC})$; $I = I_0e^{-t/RC}$	$Q_0 = CV_0$ = final charge (C); V_0 = final p.d. (V); I_0 = initial current (A); R = resistance (Ω); C = capacitance (F); t = time (s)
29	**Time constant**	$\tau = RC$	τ = time constant (s); R = resistance (Ω); C = capacitance (F); $e^{-1} = 0.368$, $e^{-2} = 0.135$
30	**Time for charge to halve**	$t_{½} = RC \ln 2 = 0.693RC$	$t_{½}$ = time for the charge to halve (s); R = resistance (Ω); C = capacitance (F)
31	**Logarithmic graph (discharge)**	Gradient of ln Q against t = $-\frac{1}{RC}$	ln Q against t is a straight line of gradient $-\frac{1}{RC}$ (discharge)

▶ CHAPTER 15 — ELECTRIC CURRENT

15.1 Conduction of Electricity

1	**Electric current**	$I = \frac{Q}{t}$	I = current (A, where A = C s^{-1}); Q = charge passing a point (C); t = time (s)
2	**Charge and number of electrons**	$Q = Ne$	N = number of electrons; e = 1.60×10^{-19} C
3	**Current and drift velocity**	$I = nAev_d$	n = free electron density (m^{-3}); A = cross-sectional area (m^2); v_d = drift velocity (m s^{-1})

15.2 Drift Velocity

4	**Current and drift velocity**	$I = nAev_d$	n = number of free charge carriers per unit volume (m^{-3}); A = cross-sectional area (m^2); e = 1.60×10^{-19} C; v_d = drift velocity (m s^{-1})
5	**Drift velocity**	$v_d = \frac{I}{nAe}$	I = current (A)
6	**Proportionality of drift velocity**	$v_d \propto I$; $v_d \propto \frac{1}{nA}$	Each holds with the other quantities constant
7	**Circular cross-sectional area**	$A = \pi r^2 = \frac{\pi d^2}{4}$	r = radius (m); d = diameter (m); for a circular cross-section
8	**Current in a semiconductor**	$I = neA(v_e + v_h)$	v_e = electron drift velocity (m s^{-1}); v_h = hole drift velocity (m s^{-1})

15.3 Current Density

9	**Current density**	$J = \frac{I}{A}$	J = current density (A m^{-2}); I = current (A); A = cross-sectional area (m^2)
10	**Current density and drift velocity**	$J = nev_d$	n = free electron density (m^{-3}); e = 1.60×10^{-19} C; v_d = drift velocity (m s^{-1})
11	**Current density and field**	$J = \sigma E$	σ = electrical conductivity (Ω^{-1} m^{-1} or S m^{-1}); E = electric field strength (V m^{-1})
12	**Field in a uniform wire**	$E = \frac{V}{L}$	V = p.d. across the wire (V); L = length of wire (m); for a uniform wire

15.4 Electric Conductivity and Resistivity

13	**Resistance of a conductor**	$R = \frac{\rho L}{A}$	R = resistance (Ω); ρ = resistivity (Ω m); L = length (m); A = cross-sectional area (m^2)
14	**Conductivity**	$\sigma = \frac{1}{\rho} = \frac{L}{RA}$	σ = conductivity (Ω^{-1} m^{-1} or S m^{-1})

15	**Resistance at constant volume**	$V = AL$; $R \propto L^2$ (or $R \propto \frac{1}{A^2}$)	V = volume (m^3), constant; L = length (m); A = cross-sectional area (m^2)
16	**Relaxation time**	$\tau = \frac{m}{ne^2\rho}$	τ = relaxation time (s); m = electron mass (9.11×10^{-31} kg); n = free electron density (m^{-3}); e = 1.60×10^{-19} C; ρ = resistivity (Ω m)
17	**Power dissipated in a resistor**	$P = I^2R = \frac{V^2}{R}$	P = power (W); I = current (A); V = p.d. (V); R = resistance (Ω); combine with $R = \frac{\rho L}{A}$ in heater-coil problems

CHAPTER 16 — DIRECT CURRENT CIRCUITS

16.1 Internal Resistance

1	**E.m.f. equation**	$E = V + Ir = I(R + r)$	E = e.m.f. (V); V = terminal p.d. (V); I = current (A); r = internal resistance (Ω); R = external resistance (Ω)
2	**Terminal p.d.**	$V = E - Ir$	A graph of V against I is a straight line with intercept E and gradient −r
3	**Internal resistance**	$r = \frac{\Delta V}{\Delta I}$; $r = \frac{E - V}{I}$	$\frac{\Delta V}{\Delta I}$ from two readings of the terminal p.d. (magnitude of the gradient of the V–I graph)
4	**Power in external resistor**	$P = I^2R = \frac{E^2R}{(R + r)^2}$	P = power in the external resistor (W)
5	**Maximum power transfer**	$R = r$; $P_{max} = \frac{E^2}{4r}$	Power in the external resistor is a maximum when R = r
6	**Power dissipated in the source**	$P_{int} = I^2r$	P_{int} = power dissipated in the internal resistance (W)
7	**Total power supplied**	$P = EI$	P = total power supplied by the source (W)

16.2 Kirchhoff's Laws

8	**Kirchhoff's first law (junction rule)**	$\Sigma I_{in} = \Sigma I_{out}$	I = current (A); applies at a junction
9	**Kirchhoff's second law (loop rule)**	$\Sigma E = \Sigma IR$	E = e.m.f. (V); I = current (A); R = resistance (Ω); applies around a closed loop
10	**Resistors in series**	$R = R_1 + R_2 + \ldots$	R = total resistance (Ω)
11	**Resistors in parallel**	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$	R = total resistance (Ω)

12	**Resistor with a voltmeter across it**	$\frac{RR_V}{R + R_V}$	R_V = finite resistance of the voltmeter (Ω); R = resistor (Ω); combined resistance when the voltmeter is connected across R

16.3 Potential Divider

13	**Potential divider output**	$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$	V_{out} = output p.d. (V); V_{in} = input p.d. (V); R_2 = resistor across which the output is taken (Ω); R_1 = other resistor (Ω)
14	**Ratio of output to input**	$\frac{V_{out}}{V_{in}} = \frac{R_2}{R_1 + R_2}$	Ratio is dimensionless
15	**Voltage ratio in series**	$V_1 : V_2 = R_1 : R_2$	For resistors in series
16	**Divider with a load**	$R_2 \rightarrow \frac{R_2 R_L}{R_2 + R_L}$	R_L = load resistance (Ω); replace R_2 by this combination in the formula
17	**Current in the divider**	$I = \frac{V_{in}}{R_1 + R_2}$; $V = IR$	I = current (A); V = p.d. across each resistor (V), using its own R

16.4 Potentiometer and Wheatstone Bridge

18	**Potentiometer balance length**	$E \propto l$; $\frac{E_1}{E_2} = \frac{l_1}{l_2}$	l = balance length (cm or m); for a uniform wire with a constant driver current
19	**E.m.f. from balance length**	$E = \frac{l}{L} V_0$	L = total wire length (same unit as l); V_0 = p.d. across the whole wire (V)
20	**P.d. across the wire**	$V_0 = IR_{wire}$	I = driver current (A); R_{wire} = resistance of the wire (Ω)
21	**Driver circuit current**	$I = \frac{E_{driver}}{R + R_{wire} + r}$	E_{driver} = e.m.f. of the driver cell (V); R = series resistance (Ω); r = internal resistance (Ω)
22	**Wheatstone bridge balance**	$\frac{P}{Q} = \frac{R}{S}$	P, Q, R, S = resistances of the four arms (Ω); at balance
23	**Slide-wire bridge**	$\frac{R}{X} = \frac{l_1}{l_2}$	X = unknown resistance (Ω); l_1, l_2 = lengths of the wire on either side of the balance point; resistance is proportional to length for a uniform wire

▶ CHAPTER 17 — MAGNETIC FIELDS

17.1 Concept of a Magnetic Field

1	**Magnetic flux density**	$B = \frac{F}{IL\sin\theta}$	B = magnetic flux density (T); F = force (N); I = current (A); L = length of conductor in the field (m); θ = angle between the conductor and B
2	**Tesla**	$1\ T = 1\ N\ A^{-1}\ m^{-1} = 1\ Wb\ m^{-2}$	T = tesla, the unit of magnetic flux density; Wb = weber
3	**Magnetic field of a straight wire**	$B = \frac{\mu_0 I}{2\pi r}$	I = current (A); r = perpendicular distance from the wire (m); $\mu_0 = 4\pi\times10^{-7}\ H\ m^{-1}$ ($T\ m\ A^{-1}$)

17.2 Force on a Moving Charge

4	**Magnetic force on a moving charge**	$F = qvB\sin\theta$	F = force (N); q = charge (C); v = speed ($m\ s^{-1}$); B = magnetic flux density (T); θ = angle between v and B
5	**Force when v ⊥ B**	$F = qvB$	Special case θ = 90°, i.e. v perpendicular to B
6	**Circular motion in a magnetic field**	$qvB = \frac{mv^2}{r}$	m = mass (kg); r = radius of the circular path (m)
7	**Radius of circular path**	$r = \frac{mv}{qB} = \frac{p}{qB}$	p = momentum ($kg\ m\ s^{-1}$); from $qvB = \frac{mv^2}{r}$
8	**Period of circular motion**	$T = \frac{2\pi m}{qB}$	T = period (s); independent of v and r
9	**Frequency of circular motion**	$f = \frac{qB}{2\pi m}$	f = frequency (Hz); independent of v and r
10	**Velocity selector**	$v = \frac{E}{B}$	E = electric field strength ($V\ m^{-1}$); B = flux density (T); v = speed ($m\ s^{-1}$)
11	**Energy gained from a p.d.**	$qV = \frac{1}{2}mv^2$	V = accelerating potential difference (V); gain in kinetic energy equals work done by the p.d.
12	**Speed after acceleration by a p.d.**	$v = \sqrt{\frac{2qV}{m}}$	v = speed ($m\ s^{-1}$); q = charge (C); V = p.d. (V); m = mass (kg)
13	**Constants for electron**	$e = 1.60\times10^{-19}$ C; $m_e = 9.11\times10^{-31}$ kg	e = elementary charge (C); m_e = electron mass (kg)

17.3 Force on a Current-Carrying Conductor

14	**Force on a current-carrying wire**	$F = BIL\sin\theta$	F = force (N); B = flux density (T); I = current (A); L = length of wire in the field (m); θ = angle between the wire and B

15	**Torque on a coil**	$\tau = NBIA\sin\varphi$	τ = torque (N m); N = number of turns; A = area of the coil (m^2); φ = angle between the normal to the coil and B
16	**Wire balanced against its weight**	$BIL = mg$	m = mass of wire (kg); g = acceleration of free fall ($m\ s^{-2}$); B perpendicular to the wire

17.4 Magnetic Fields Due to Currents

17	**Field of a long straight wire**	$B = \frac{\mu_0 I}{2\pi r}$	B = flux density (T); I = current (A); r = perpendicular distance from the wire (m)
18	**Field at centre of a circular coil**	$B = \frac{\mu_0 NI}{2r}$	N = number of turns; r = radius of the coil (m); I = current (A)
19	**Field inside a long solenoid**	$B = \mu_0 nI = \frac{\mu_0 NI}{l}$	n = number of turns per metre (m^{-1}); N = total number of turns; l = length of solenoid (m)
20	**Field of a toroid**	$B = \frac{\mu_0 NI}{2\pi r}$	r = mean radius of the ring (m); N = number of turns; I = current (A)
21	**Permeability of free space**	$\mu_0 = 4\pi\times10^{-7}\ H\ m^{-1} = 1.26\times10^{-6}\ H\ m^{-1}$	μ_0 = permeability of free space (H m^{-1})

17.5 Force Between Two Current-Carrying Conductors

22	**Force per unit length between wires**	$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$	$\frac{F}{L}$ = force per unit length (N m^{-1}); I_1 and I_2 = currents (A); d = separation (m)
23	**Definition check (ampere)**	$\frac{F}{L} = \frac{4\pi\times10^{-7}}{2\pi} = 2\times10^{-7}\ N\ m^{-1}$	For $I_1 = I_2 = 1$ A and d = 1 m
24	**Force on a length of wire**	$F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$	l = length of one wire (m); I_1 and I_2 = currents (A); d = separation (m)

17.6 Determination of the Ratio e/m

25	**Balanced fields (velocity)**	$eE = evB;\ v = \frac{E}{B}$	e = electron charge (C); E = electric field strength (V m^{-1}); B = flux density (T); v = speed (m s^{-1})
26	**Magnetic deflection**	$r = \frac{mv}{eB};\ \frac{e}{m} = \frac{v}{rB}$	r = radius of circular path (m); m = electron mass (kg)
27	**Accelerating p.d.**	$eV = \frac{1}{2}mv^2;\ v = \sqrt{\frac{2eV}{m}}$	V = accelerating p.d. (V)
28	**Combined expression for e/m**	$\frac{e}{m} = 2V/(B^2r^2)$	$\frac{e}{m}$ = specific charge of the electron (C kg^{-1})

29	**Accepted value of e/m**	$\frac{e}{m} = 1.76\times10^{11}\ \text{C kg}^{-1}$	Accepted value of the specific charge (C kg^{-1})

17.7 Hall Effect

30	**Hall voltage**	$V_H = \frac{BI}{nte}$	V_H = Hall voltage (V); B = flux density (T); I = current (A); n = number density of carriers (m^{-3}); t = thickness of the slab in the direction of B (m); e = 1.60×10^{-19} C
31	**Balance of forces**	$\frac{eV_H}{d} = evB$	d = width of the slab between the Hall faces (m); v = drift velocity (m s^{-1})
32	**Current in terms of drift velocity**	$I = nevtd$	Substituting into the force balance gives $V_H = \frac{BI}{nte}$
33	**Drift velocity**	$v = \frac{I}{nAe}$	A = cross-sectional area (m^2); n = number density of carriers (m^{-3}); e = 1.60×10^{-19} C

CHAPTER 18 — ELECTROMAGNETIC INDUCTION

18.1 Magnetic Flux

1	**Magnetic flux**	$\Phi = BA\cos\theta$	Φ = magnetic flux (Wb = T m^2); B = magnetic flux density (T); A = area (m^2); θ = angle between B and the normal to the plane
2	**Flux linkage**	$N\Phi = NBA\cos\theta$	NΦ = flux linkage (Wb-turns); N = number of turns
3	**Maximum flux**	$\Phi = BA$ at $\theta = 0°$	Plane perpendicular to B
4	**Zero flux**	$\Phi = 0$ at $\theta = 90°$	Plane parallel to B
5	**Area of circular coil**	$A = \pi r^2$	A = area (m^2); r = radius (m)
6	**Area of square coil**	$A = (\text{side})^2$	A = area (m^2); side in m

18.2 Faraday's Law and Lenz's Law

7	**Faraday's law**	$\varepsilon = -N\frac{d\Phi}{dt} = -\frac{d(N\Phi)}{dt}$	ε = induced emf (V); N = number of turns; Φ = magnetic flux (Wb); t = time (s); the minus sign represents Lenz's law
8	**Faraday's law, uniform change**	$\lvert\varepsilon\rvert = \frac{N\Delta\Phi}{\Delta t}$	ε = induced emf (V); N = number of turns; ΔΦ = change in flux (Wb); Δt = time interval (s); for a uniform change

9	**Emf induced in a moving rod**	$\varepsilon = BLv$	ε = emf (V); B = magnetic flux density (T); L = length of rod (m); v = velocity of rod (m s^{-1}); B, L and v mutually perpendicular
10	**Emf with v at angle to B**	$\varepsilon = BLv \sin\theta$	θ = angle between v and B; other symbols as above
11	**Rod in a closed circuit**	$I = \frac{BLv}{R}$; $F = BIL = \frac{B^2L^2v}{R}$; $P = Fv = \frac{\varepsilon^2}{R}$	I = current (A); R = resistance of circuit (Ω); F = force (N); P = power (W); B, L, v as above; ε = BLv
12	**Emf in a rotating coil**	$\varepsilon = NBA\omega \sin\omega t$; $\varepsilon_0 = NBA\omega$	ε = induced emf (V); ε_0 = peak emf (V); N = number of turns; B = magnetic flux density (T); A = area (m^2); ω = angular speed (rad s^{-1}); uniform field

18.3 Self Induction

13	**Self-induced emf**	$\varepsilon = -L\frac{dI}{dt}$	ε = induced emf (V); L = self-inductance (H); $\frac{dI}{dt}$ = rate of change of current (A s^{-1})
14	**Self-induced emf, uniform change**	$\lvert\varepsilon\rvert = L\frac{\Delta I}{\Delta t}$	ΔI = change in current (A); Δt = time interval (s); for a uniform change
15	**Self-inductance**	$L = \frac{N\Phi}{I}$	L = self-inductance (H = Wb A^{-1} = V s A^{-1}); N = number of turns; Φ = flux (Wb); I = current (A)
16	**Inductance of air-cored solenoid**	$L = \frac{\mu_0 N^2 A}{l}$	L = self-inductance (H); N = number of turns; A = cross-sectional area (m^2); l = length (m); $\mu_0 = 4\pi\times10^{-7}$ H m^{-1}; long air-cored solenoid
17	**Inductance with a core**	$L = \frac{\mu_r\mu_0 N^2 A}{l}$	μ_r = relative permeability of the core (no unit); other symbols as above

18.4 Energy Stored in an Inductor

18	**Energy stored in an inductor**	$E = \frac{1}{2}LI^2$	E = energy (J); L = self-inductance (H); I = current (A)
19	**Energy in terms of flux linkage**	$E = \frac{N\Phi I}{2}$	N = number of turns; Φ = flux (Wb); I = current (A); since NΦ = LI
20	**Magnetic energy density**	$u = \frac{B^2}{2\mu_0}$	u = energy density (J m^{-3}); B = magnetic flux density (T); $\mu_0 = 4\pi\times10^{-7}$ H m^{-1}
21	**Average power released**	$P = \frac{E}{t}$	P = average power (W); E = energy (J); t = time (s)

18.5 Mutual Induction

22	**Mutually induced emf**	$\varepsilon_2 = -M\frac{dI_1}{dt}$	ε_2 = emf induced in secondary (V); M = mutual inductance (H); $\frac{dI_1}{dt}$ = rate of change of primary current (A s^{-1})
23	**Mutually induced emf, uniform change**	$\lvert\varepsilon_2\rvert = M\frac{\Delta I_1}{\Delta t}$	ΔI_1 = change in primary current (A); Δt = time interval (s); for a uniform change
24	**Mutual inductance**	$N_2\Phi_{21} = MI_1$	N_2 = number of turns of secondary; Φ_{21} = flux in secondary due to primary (Wb); M = mutual inductance (H = Wb A^{-1}); I_1 = primary current (A)
25	**Ideal transformer**	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$; $V_pI_p = V_sI_s$	V = voltage (V); N = number of turns; I = current (A); subscripts p = primary, s = secondary
26	**Coupling coefficient**	$M = k\sqrt{L_1L_2}$	M = mutual inductance (H); k = coupling coefficient, $0 \le k \le 1$; L_1, L_2 = self-inductances of the two coils (H)

CHAPTER 19 — ALTERNATING CURRENT CIRCUITS

19.1 AC Through a Pure Resistor

1	**Current and voltage**	$I = I_0 \sin\omega t$; $V = V_0 \sin\omega t$	I = current (A); V = voltage (V); I_0, V_0 = peak values; $\omega = 2\pi f$ (rad s^{-1}); f = frequency (Hz); t = time (s)
2	**Period**	$T = \frac{1}{f}$	T = period (s); f = frequency (Hz)
3	**r.m.s. values**	$I_{rms} = \frac{I_0}{\sqrt{2}}$; $V_{rms} = \frac{V_0}{\sqrt{2}}$	I_{rms} = r.m.s. current (A); V_{rms} = r.m.s. voltage (V); I_0, V_0 = peak values; sinusoidal waveform only
4	**Peak current from r.m.s.**	$I_0 = \sqrt{2}\, I_{rms} \approx 1.41\, I_{rms}$	I_0 = peak current (A); I_{rms} = r.m.s. current (A)
5	**Mean power**	$P = I_{rms}^2R = \frac{V_{rms}^2}{R} = I_{rms}V_{rms} = \frac{1}{2}I_0^2R = \frac{1}{2}V_0I_0$	P = mean power (W); R = resistance (Ω); I_{rms}, V_{rms} = r.m.s. current (A) and voltage (V); I_0, V_0 = peak values
6	**Peak power**	$P_{peak} = I_0^2R = 2P_{mean}$	P_{peak} = peak power (W); P_{mean} = mean power (W); I_0 = peak current (A); R = resistance (Ω)
7	**Energy dissipated**	$E = P_{mean}\, t$	E = energy (J); P_{mean} = mean power (W); t = time (s)

19.2 AC Through a Pure Inductor

8	**Inductive reactance**	$X_L = \omega L = 2\pi fL$	X_L = inductive reactance (Ω); ω = angular frequency (rad s^{-1}); f = frequency (Hz); L = inductance (H)
9	**Current in an inductor**	$I_{rms} = \frac{V_{rms}}{X_L}$; $I_0 = \frac{V_0}{X_L}$	I_{rms}, V_{rms} = r.m.s. current (A) and voltage (V); I_0, V_0 = peak values; X_L = inductive reactance (Ω)
10	**Mean power in a pure inductor**	$P = V_{rms} I_{rms} \cos\varphi = 0$	P = mean power (W); φ = phase angle = 90°; V_{rms}, I_{rms} = r.m.s. voltage (V) and current (A)

19.3 AC Through a Pure Capacitor

11	**Capacitive reactance**	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$	X_C = capacitive reactance (Ω); ω = angular frequency (rad s^{-1}); f = frequency (Hz); C = capacitance (F)
12	**Current in a capacitor (r.m.s.)**	$I_{rms} = \frac{V_{rms}}{X_C}$	I_{rms} = r.m.s. current (A); V_{rms} = r.m.s. voltage (V); X_C = capacitive reactance (Ω)
13	**Current in a capacitor (peak)**	$I_0 = \frac{V_0}{X_C} = \omega C V_0$	I_0 = peak current (A); V_0 = peak voltage (V); X_C = capacitive reactance (Ω)
14	**Reactance and frequency**	$X_C \propto \frac{1}{f}$	X_C is inversely proportional to f; a graph of X_C against f is a decreasing curve (rectangular hyperbola) approaching zero at high frequency
15	**Mean power in a capacitor**	$P = V_{rms} I_{rms} \cos\varphi = 0$	P = mean power (W); φ = phase angle between V and I = 90° for a pure capacitor, so cosφ = 0

19.4 R-C and R-L Series Circuits

16	**R-L series impedance**	$Z = \sqrt{R^2 + X_L^2}$	Z = impedance (Ω); R = resistance (Ω); X_L = inductive reactance (Ω)
17	**R-L series phase angle**	$\tan\varphi = \frac{X_L}{R}$	φ = phase angle between voltage and current (rad or °)
18	**Inductive reactance**	$X_L = 2\pi fL$	X_L = inductive reactance (Ω); f = frequency (Hz); L = inductance (H)
19	**R-C series impedance**	$Z = \sqrt{R^2 + X_C^2}$	Z = impedance (Ω); R = resistance (Ω); X_C = capacitive reactance (Ω)
20	**R-C series phase angle**	$\tan\varphi = \frac{X_C}{R}$	φ = phase angle between voltage and current (rad or °)
21	**Capacitive reactance**	$X_C = \frac{1}{2\pi fC}$	X_C = capacitive reactance (Ω); f = frequency (Hz); C = capacitance (F)

22	**Supply voltage and current**	$V_{rms} = I_{rms} Z$	V_{rms} = r.m.s. supply voltage (V); I_{rms} = r.m.s. current (A); Z = impedance (Ω)
23	**Voltage relation (phasor addition)**	$V^2 = V_R^2 + V_L^2$ (or V_C^2); $V_R = IR$, $V_L = IX_L$, $V_C = IX_C$	V = supply voltage (V); V_R, V_L, V_C = voltages across R, L, C (V); I = current (A)
24	**Power factor**	$\cos\varphi = \frac{R}{Z}$	cosφ = power factor (no unit); R = resistance (Ω); Z = impedance (Ω)
25	**Mean power**	$P = V_{rms} I_{rms} \cos\varphi = I_{rms}^2 R$	P = mean power (W); V_{rms}, I_{rms} = r.m.s. voltage (V) and current (A); R = resistance (Ω)